Supplemental materials

Title: Translational modeling of gut microbiome-mediated drug metabolism: a case example of sulfasalazine

Author: Lisa Cheng[1,2,†], Jingwei Cai[1,†], Alexis Auster[1], Sungjoon Cho[1], Mary Keir[3], Horace Rhee[4], Bennett Kapili[5], Dari Shalon[5], Miaoran Ning[1,*]

Affiliation: 1. Drug Metabolism and Pharmacokinetics, Genentech, Inc., South San Francisco, CA, USA 94080; 2. Faculty of Pharmaceutical Sciences, The University of British Columbia, Vancouver, BC, V6T 1Z3; 3. Human Pathobiology and OMNI Reverse Translation, Genentech, Inc., South San Francisco, CA, USA 94080; 4. Early Clinical Development, OMNI, Genentech, Inc., South San Francisco, CA, USA 94080; 5. Envivo Bio, Inc., San Francisco, CA, USA 94107

***Corresponding author:** Miaoran Ning, Drug Metabolism and Pharmacokinetics, Genentech, Inc., 1 DNA Way, South San Francisco, CA, 94080, Phone: 650-822-4207 Email: ning.miaoran@gene.com

[†] These authors contributed equally to this work

Supplementary materials and methods

Chemicals and Reagents

LC-MS grade water, acetonitrile (ACN), ammonium acetate, and methanol were obtained from ThermoFisher Scientific Inc. (Rockford, IL). Sulfasalazine, mesalamine, sulfasalazine-d4 and mesalamine-d3 were purchased from Toronto Research Chemicals Inc (North York, ON, Canada).

Time-dependent sulfasalazine kinetic evaluation

Fresh fecal samples from two healthy donors were previously pooled under anaerobic condition and frozen in an ultra-low temperature (-80°C) freezer before experimental use. Pooled feces was then transferred into an anaerobic environment to thaw before preparing 100 times diluted fecal slurries in culture medium. Next, 98 µL aliquots of the fecal slurries were dosed in triplicate with 2 µL of 5 mM sulfasalazine stock in a 96-well plate to yield a working concentration of 100 µM. The incubation plates were placed in a shaking incubator at 37°C. The dosed samples were then crashed with 200 µL ice-cold ACN at 0, 5, 10, 15, 30, 60, 90, 120, and 240 minutes, and stored in an ultra-low temperature freezer for metabolite extraction and LC-MS analysis. The crashed time point plates were centrifuged at 3750 rpm for 20 minutes (Beckman Coulter Allegra X 12R) and 100 µL supernatants were transferred to an LC-MS autosampler compatible plate. For quantification, a standard curve was included ranging from: 0-100 µM. 100 µL of the supernatants were then dried down with nitrogen and reconstituted in 180 µL of 10 mM ammonium acetate water containing isotope-labeled sulfasalazine-d4 (0.1 µM) and mesalamine-d3 (1 µM) as internal standards (IS). The final plate was prepared by diluting the

samples 10-fold using 10 mM ammonium acetate water with IS to determine the linear range of sulfasalazine turnover over time.

LC-MS method

Final plates were analyzed on a Q-Trap 6500+ mass spectrometer (SCIEX LLC, Framingham, MA) coupling a Shimadzu Nexera X2 ultra-performance liquid chromatography (UPLC) (Kyoto, Japan), with a Kinetex C18 column (2.6 µm, 100Å, 30 x 2.1 mm) (Torrance, CA). The steady flow rate was 0.6 mL/min, with the initial condition of 1% organic phase B (methanol) and 99% aqueous phase A (10 mM ammonium acetate water) for 0.5 min. From 0.5 min to 2 min, the mobile phase B ramped up to 90% and held until 2.5 min, then ramped down to 1% within 0.3 min, and held at 1% for an additional 1 min. The total method was 3.8 min. Mass spectrometry was performed with multiple reaction monitoring (MRM) methods. All LC-MS/MS data were processed with MultiQuant™ Software (SCIEX). MRM transitions (Q1/Q3) for all test drugs are listed in the Supplemental Table S1.

Gut luminal sample and feces collection in healthy subjects

Briefly, gut luminal samples were collected in a sampling capsule (CapScan®, Envivo Bio Inc), a single-use ingested device that collects fluid from the human intestines for *ex vivo* analysis. The capsules were administered in the morning after an overnight fast. The sampling capsule collected luminal samples from different regions of the intestinal tract over the course of transit time during normal peristalsis. The collection tube released from capsule was passed into the toilet during a bowel movement and stored at 4°C. Feces were collected in the first bowel movement after ingestion of the sampling capsule and immediately stored at 4°C. All samples

were aliquoted into analysis tubes within 24 hours, which were then frozen at -80°C until further analysis.

Subject inclusion and exclusion criteria: Subjects met all of the following criteria: 18-70 years old; ASA classification 1 or 2; nonpregnant; agree to remain abstinent; fluent in English; and is willing and able to comply with study requirements. Subjects with known or suspected conditions that may interrupt microflora, impact GI function, or potentially pose a health risk to the subject while involved in the study were excluded.

Bacterial count

Bacterial count was estimated in 9 of the 10 healthy subjects. 20 µl of ZymoBIOMICS Spike-in Control II (Zymo Research, cat. No. D6321) were added to samples immediately prior to DNA extraction. After sequencing, samples with total CosmosID abundance scores ≥1000 were retained for further analysis. First, taxon abundance scores were transformed to relative abundances, then an ordinary least squares model with formula spike-in taxon relative abundance and known spiked cell count was fit for each sample separately using the lm function in the R stats package (v.4.2.1). Total endogenous community cell counts were estimated by subtracting the model intercept coefficients from the total non-spike community relative abundances and dividing by the model slope coefficients. Cell count estimates were normalized to sample input volumes and multiplied by sample dilution factors to yield dilution-corrected microbial count estimates expressed in cells/µL.

Literature PK collection and digitization

A comprehensive literature search was conducted for clinical studies where sulfasalazine, sulfapyridine, or mesalamine was administered. Only studies of healthy adult volunteers with

reported concentration-time profiles of sulfasalazine, sulfapyridine, and/or mesalamine were included. PK profiles that reported total metabolite levels (e.g., sulfapyridine and acetylated sulfapyridine) were excluded. PK profiles were digitized using WebPlotDigitizer V4.6. Noncompartmental analysis was performed using Phoenix® WinNonlin® when a concentration-time profile was published without corresponding exposure parameters.

Dog cassette pharmacokinetic study

A dog PK study was designed to estimate regional effective permeability (P_{eff}) of sulfapyridine and mesalamine. Three male beagle dogs were given four cassette doses of sulfapyridine and mesalamine in solution once a week in the follow order: (1) intravenous (IV) bolus dose of 3 mg/kg sulfapyridine and 2 mg/kg mesalamine (2) oral dose of 15 mg/kg sulfapyridine and 10 mg/kg mesalamine (3) intraileum dose of 15 mg/kg sulfapyridine and 10 mg/kg mesalamine; (4) intracolon dose of 15 mg/kg sulfapyridine and 10 mg/kg mesalamine. 30% HP β-CD, pH=9.4 or 5% DMSO/10% Solutol/85% water was used as vehicle for IV or non-IV formulations respectively. Plasma samples were collected and analyzed for both sulfapyridine and mesalamine concentrations at 0.05, 0.167, 0.33, 0.5, 0.75, 1, 2, 4, 6, 8, 12, 24 hour post IV dose or 0.25, 0.5, 1, 2, 4, 6, 8, 10, 12, 24 hour post non-IV doses. Noncompartmental analysis was performed using Phoenix® WinNonlin® (Model 200 202; Version 6.4.1; Certara USA; Princeton, NJ). Dog 2 data from intraileum dose phase and Dog 1 data from intracolon dose phase were excluded from P_{eff} estimation due to abnormal absorption faster than that in the same dog during all other non-IV dosing phases and that in other dogs from the same dosing phase. The phenomena were suspected to be artifact of intraluminal dosing other than difference in the dog regional permeability.

Estimation of dog regional P_{eff}

Dog PBPK model for sulfapyridine and mesalamine were built with Simcyp Animal simulator V22R1. The PBPK input parameters were listed in supplemental Table S4. IV PK were used to inform systemic PK parameters for both molecules. Then PO arm was used to optimize PK at oral doses. Systemic clearance of sulfapyridine was decreased for non-IV doses based on >100% bioavailability and mild increase in terminal half-life. Intestinal metabolism was added to mesalamine PBPK model to recover observed bioavailability. Intracolon dosing arm were simulated using intracolon dosing route. Colon P_{eff} values for sulfapyridine and mesalamine were optimized to capture observed PK profile after intracolon dosing. Intraileum dosing option was not available in Simcyp® animal simulator. The dosing arm was simulated as enteric-coated tablet formulation with trigger pH of 6 and adjusting dog GI luminal pH to be 5 for segments before ileum to mimic immediate ileum dosing. When comparing simulated PK with observed data, simulated time points were offset by 2.64 hour to subtract out GI transit time to ileum. Ileum P_{eff} values for both molecules were then optimized with fixed optimized colon P_{eff} values to capture observed PK after intraileum dosing. Colon transit time was decreased from 7.45 hour to 3 hour for this arm to recover terminal half-life of mesalamine. Jejunum P_{eff} values were optimized last with fixed ileum and colon P_{eff} values from above to capture observed oral PK absorption.

Supplementary Tables

Table S1. MRM transitions for analytes of interest

Compound	**Q1/Q3**	**Mode**
Mesalamine	152.0/108.0	-
Mesalamine-d3	154.0/110.0	-
Sulfapyridine	205.2/156.0	+
Sulfasalazine	399.1/223.0	+
Sulfasalazine-d4	403.1/223.0	+

Table S2. Bacterial count in colon and stool samples in 9 healthy subjects.

Subject	Colon bacterial count cells/µL	Stool bacterial count cells/µL	Ratio Colon/Stool
Subject 1	1.93×10^{7}	2.10×10^{7}	0.92
Subject 2	1.97×10^{6}	5.36×10^{6}	0.37
Subject 3	1.06×10^{7}	4.34×10^{7}	0.24
Subject 4	2.27×10^{7}	3.40×10^{7}	0.67
Subject 5	1.67×10^{6}	2.54×10^{7}	0.07
Subject 6	1.75×10^{7}	1.92×10^{7}	0.91
Subject 7	6.63×10^{7}	1.14×10^{8}	0.58
Subject 8	NA	2.44×10^{7}	NA
Subject 9	NA	9.04×10^{6}	NA
Geometric mean	1.08×10^{7}	2.33×10^{7}	0.47
CV%	102	93	56

Table S3. Summary of observed and predicted PK of sulfasalazine, sulfapyridine, and mesalamine when dosed alone.

Study	N (ratio of females)	Age (years)	Weight (kg)	ABCGG2 Genotype	NAT2 Genotype	Dose	Analyte	Exposure Parameters			
									C_{max} (µg/mL)	T_{max} (h)	AUC_{o-t} (µg/mL*h)
Model: Sulfasalazine											
Kusuhara et al. 2012[1]	8 (0)	NR	NR	C/C	NR	2000 mg Sulfasalazine (PO)	Sulfasalazine	Observed	13.16±1.16	NA	117.80±14.20
								Predicted	16.55	4.10	188.56
								Fold-Error	0.80	NA	0.62
Kusuhara et al. 2012[1]	8 (0)	NR	NR	C/C	NR	2000 mg Sulfasalazine + 2000 mg Curcumin (PO)	Sulfasalazine	Observed	49.00±5.80	NA	380.00±54.00
								Predicted	51.64	4.70	514.40
								Fold-Error	0.95	NA	0.74
Yamasaki et al. 2008[2]	12 (0)	20 to 24	49.5 to 79.7	C/C	NR	2000 mg Sulfasalazine (PO)	Sulfasalazine	Observed	15.40±5.20	4..30±0.90	171.00±85.00
								Predicted	16.55	4.10	202.42
								Fold-Error	0.93	1.05	0.84
Yamasaki et al. 2008[2]	16 (0)	20 to 24	49.5 to 79.7	C/A	NR	2000 mg Sulfasalazine (PO)	Sulfasalazine	Observed	26.40±16.10	4.80±1.60	330.00±194.00
								Predicted	22.58	4.30	279.91
								Fold-Error	1.17	1.12	1.18
Yamasaki et al. 2008	9 (0)	20 to 24	49.5 to 79.7	A/A	NR	2000 mg Sulfasalazine (PO)	Sulfasalazine	Observed	40.70±13.50	3.90±2.20	592.00±275.00
								Predicted	35.16	4.60	440.87
								Fold-Error	1.16	0.85	1.34
Schroder and Campbell 1972[2]	9 (0)	21 to 29	61 to 78	NR	NR	4000 mg Sulfasalazine (PO)	Sulfasalazine	Observed *	27.47±12.31	4.50±1.41	229.26±146.76
								Predicted	17.67	3.80	188.01
								Fold-Error	1.55	1.18	1.22
Das et al. 1979[3]	3 (0)	NR	NR	NR	NR	2000 mg Sulfasalazine (PO)	Sulfasalazine	Observed *	22.13±0.01	5.00	227.57±0.11
								Predicted	13.40	4.20	156.68
								Fold-Error	1.65	1.19	1.45

Model: Sulfapyridine											
Vree et al. 1980[4]	1 (NR)	NR	NR	NR	Slow Acetylator	500 mg Sulfapyridine (PO)	Sulfapyridine	Observed	15.20	2.82	164.00
								Predicted	10.57	1.10	195.37
								Fold-Error	1.44	2.56	0.84
Vree et al. 1980[4]	1 (NR)	NR	NR	NR	Rapid Acetylator	588 mg Sulfapyridine (PO)	Sulfapyridine	Observed	9.69	1.66	81.10
								Predicted	9.94	1.00	87.97
								Fold-Error	0.97	1.66	0.92
Model: Mesalamine											
Norlander et al. 1989[5]	12 (0.5)	20 to 43	NR	NR	NR	500 mg Mesalamine (PO)	Mesalamine	Observed	1.20	6.50	3.00
								Predicted	0.63	5.28	3.64
								Fold-Error	1.90	1.23	0.82
Yu et al. 1990[6]	23 (0)	27.9±5.4	75.75±9.34	NR	NR	1000 mg Mesalamine (PO)	Mesalamine	Observed	14.72	0.98	28.67
								Predicted	12.68	0.85	27.01
								Fold-Error	1.16	1.15	1.06

* denotes studies where the pharmacokinetic properties were calculated using an NCA analysis of the digitized concentration-time profile

Observed values are reported as mean or mean ± SD. Fold-error analysis of PK parameters were calculated as a ratio of observed to predicted. $AUC_{0\text{-}t}$: area under the concentration-time curve from time 0 h to t; C_{max}: maximum plasma concentration; IV: dose administered intravenously; NA: not available; NR: not reported; PO: dose administered by mouth; T_{max}: time when C_{max} is achieved.

Table S4. Dog PBPK parameters for sulfapyridine and mesalamine.

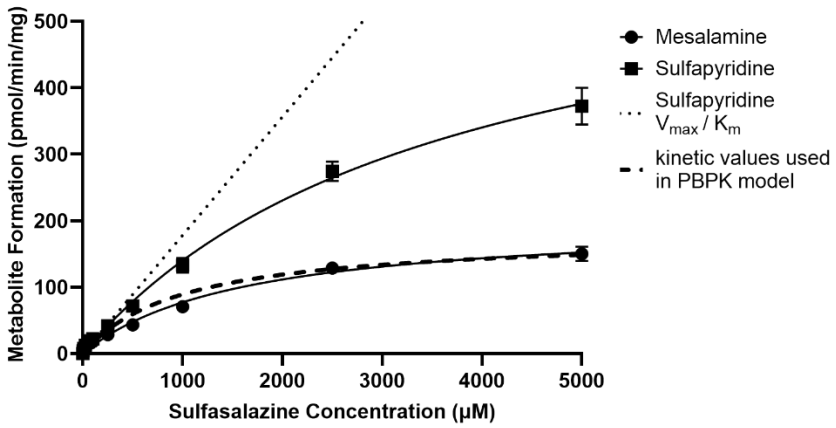
Parameter	Value	Methods/References
Sulfapyridine		
MW, g/mol	249.29	
logP	0.35	
pK_a	8.43	Monoprotic Acid
$f_{u,p}$	0.3	Measured in house
B:P ratio	0.55	Default for acid
Absorption model	ADAM model solution formulation	
P_{eff}, 10^{-4} cm/s Duodenum,jejunum,ileum,colon	5,5,2.5,2.5	Optimized to fit observed PO, intraileum and intracolon PK
Distribution	Full PBPK model	
V_{ss}, L/kg	1.25 (Method 2)	Observed V_{ss}
Tissue Kp scalar	0.74	To match observed Vss
Elimination		
CL_{iv}, mL/min	96.2 (3 mg/kg) / 57.3 (15 mg/kg)	Observed value from IV PK in 3 mg/kg; decrease 1.67x for 15 mg/kg based on apparent bioavailability and increase in terminal half-life at higher dose
Mesalamine		
MW, g/mol	153.14	
logP	0.98	
pK_{a1}, pK_{a2}	2.3, 5.69	Ampholyte
$f_{u,p}$	0.45	Measured in house
B:P ratio	0.55	Default for acid
Absorption model	ADAM model solution formulation	
P_{eff}, 10^{-4} cm/s Duodenum,jejunum,ileum,colon	5,5,0.2,0.2	Optimized to fit observed PO, intraileum and intracolon PK
Distribution	Full PBPK model	
V_{ss}, L/kg	0.44 (Method 2)	Observed V_{ss}
Tissue Kp scalar	2.08	To match observed Vss
Elimination		
$Cl_{int,liver\ microsome}$ µL/min/mg microsome protein	9.07	Retro-calculated from IV CL
$Cl_{int,intestinal\ microsome}$ µL/min/mg microsome protein	6.99	Optimized to recover solution oral bioavailability

Table S5. Comparison between the PBPK model prediction and observed PK in dog after oral, intraileum or intracolon cassette dose of mesalamine (10 mg/kg) and sulfapyridine (15 mg/kg).

Dosing Route		Mesalamine 10 mg/kg			Sulfapyridine 15 mg/kg		
		$AUC_{0\text{-last}}$ h×μmol/L	C_{max} μmol/L	T_{max} h	$AUC_{0\text{-last}}$ h×μmol/L	C_{max} μmol/L	T_{max} h
Oral	Observed	137	56.6	0.67	163	36.7	1.0
	Predicted	179	66.0	0.80	157	33.4	0.90
	Pred./Obs.	1.31	1.17	1.20	0.96	0.91	0.90
Intraileum*	Observed	66.8	13.6	2.0	186	34.3	1.0
	Predicted	72.1	13.5	1.7	141	28.5	0.86
	Pred./Obs.	1.08	0.99	0.87	0.76	0.83	0.86
Intracolon*	Observed	108	18.0	2.0	205	34.9	2.0
	Predicted	102	13.6	2.5	159	28.7	1.1
	Pred./Obs.	0.95	0.76	1.23	0.78	0.82	0.55

* The same three dogs were dosed in each dosing phase. Reported PK from n=2 dogs after excluding outlier of the group.

Supplementary Figures



	Mesalamine	Sulfapyridine	*Kinetic values used in PBPK model
V_{max} (pmol/min/mg feces)	200.9	650.5	178.3
K_m (μM)	1605	3648	1000
CL_{int} (V_{max}/K_m, μL/min/mg feces)	0.125	0.178	0.178

**Sulfapyridine V_{max} and K_m value both scaled down 3.6-fold to maintain same CL_{int} value while K_m=1000 μM.*

Figure S1. *In vitro* kinetics of sulfasalazine conversion to sulfapyridine and mesalamine using pooled fecal samples from four healthy donors.

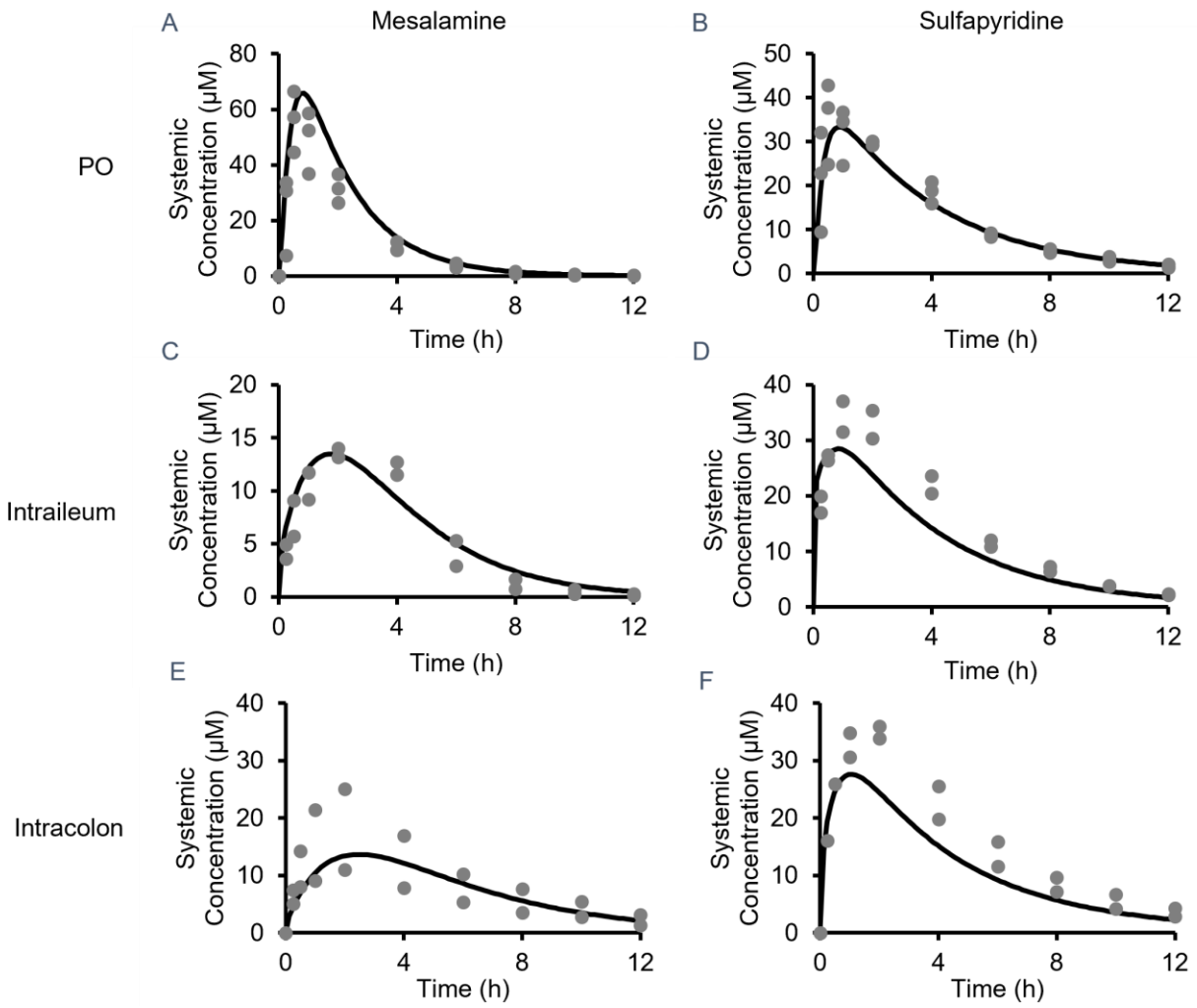


Figure S2. Observed (solid grey dots) vs. predicted (solid line) concentration-time profile in dog after cassette dosing of 10 mg/kg mesalamine (A, C, E) and 15 mg/kg sulfapyridine (B, D, F) via oral (A, B), intra-ileum (C, D) or intra-colon (E, F) solution dosing.

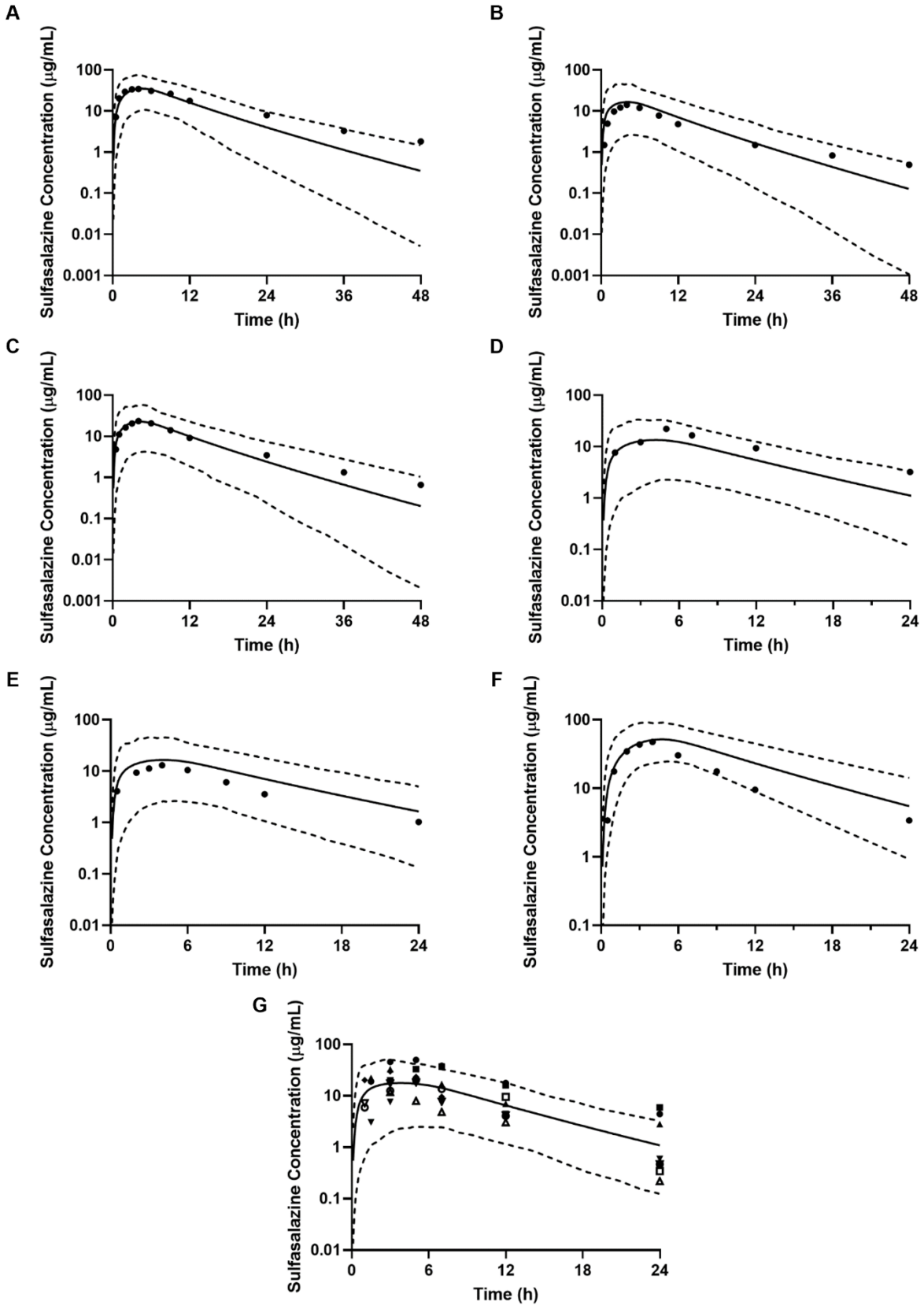

A
B
C
D
E
F
G
Sulfasalazine Concentration (μg/mL)
Time (h)
100
10
1
0.1
0.01
0.001
0
12
24
36
48
6
18

Figure S3. Simulated (solid black line) and mean observed (closed circles) concentration-time profiles of sulfasalazine when dosed with sulfasalazine. Dashed black lines represent the simulated 5th to 95th percentile of the virtual population. (A-C) Yamasaki et al. 2008[7] genotyped healthy Japanese volunteers as the (A) 421A/C (n=9), (B) 421A/A (n=16), and (C) 421C/C (n=12) ABCG2 genotype who received a 2 g dose of sulfasalazine vs. corresponding PBPK simulation. D) Healthy male volunteers (n=3) received a 2 g dose of sulfasalazine vs. corresponding PBPK simulation (Das et al. 1979[3]). E) Healthy male Japanese volunteers (n=8) with the ABCG2 C/C genotype were given a 2 g dose of sulfasalazine vs. corresponding PBPK simulation (Kusuhara et al. 2012[1]). F) Healthy male Japanese volunteers (n=8) were given a 2 g dose of sulfasalazine and 2 g of curcumin vs. corresponding PBPK simulation. G) Healthy volunteers (n=9) orally administered 4 g of sulfasalazine. (Schröder and Campbell 1972[8]) vs. corresponding PBPK simulation.

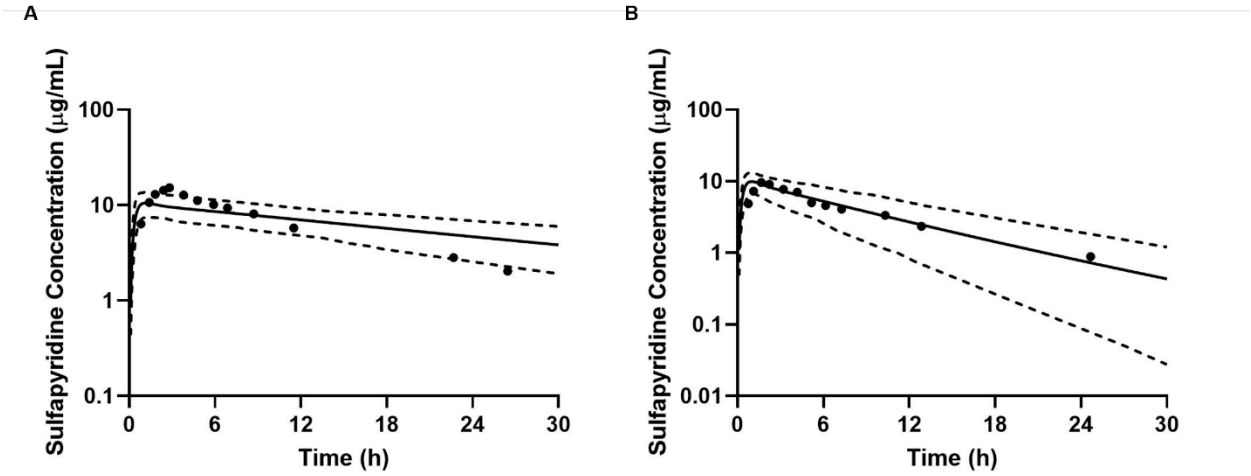


Figure S4. Simulated (solid black line) and observed (closed circles) concentration-time profiles of sulfapyridine for the verification of the sulfapyridine physiologically-based pharmacokinetic model. Dashed black lines represent the simulated 5th to 95th percentile of the virtual

population. One slow acetylator (A) and one fast acetylator (B) orally administered 505 and 588 mg of sulfapyridine (Vree et al. 1980[4]).

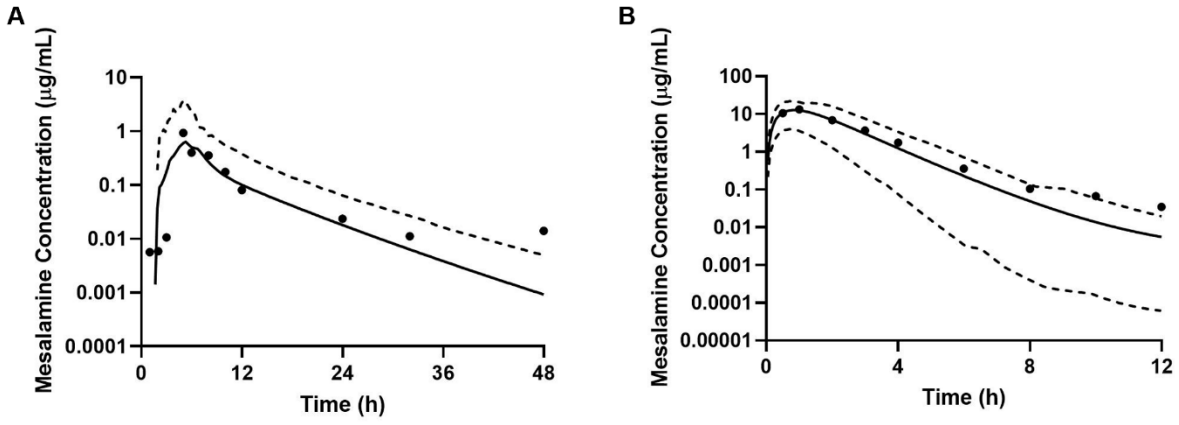


Figure S5. Simulated (solid black line) and observed (closed circles) concentration-time profiles of mesalamine. Dashed black lines represent the simulated 5th to 95th percentile of the virtual population. (A) Healthy volunteers (n=12) were given two enteric-coated formulations totaling 500 mg dose of mesalamine vs. corresponding PBPK simulation. (Norlander et al. 1989[5]) (B) Volunteers (n=23) received an oral dose of 1000 mg mesalamine suspension vs. corresponding PBPK simulation. (Yu et al. 1990[6])

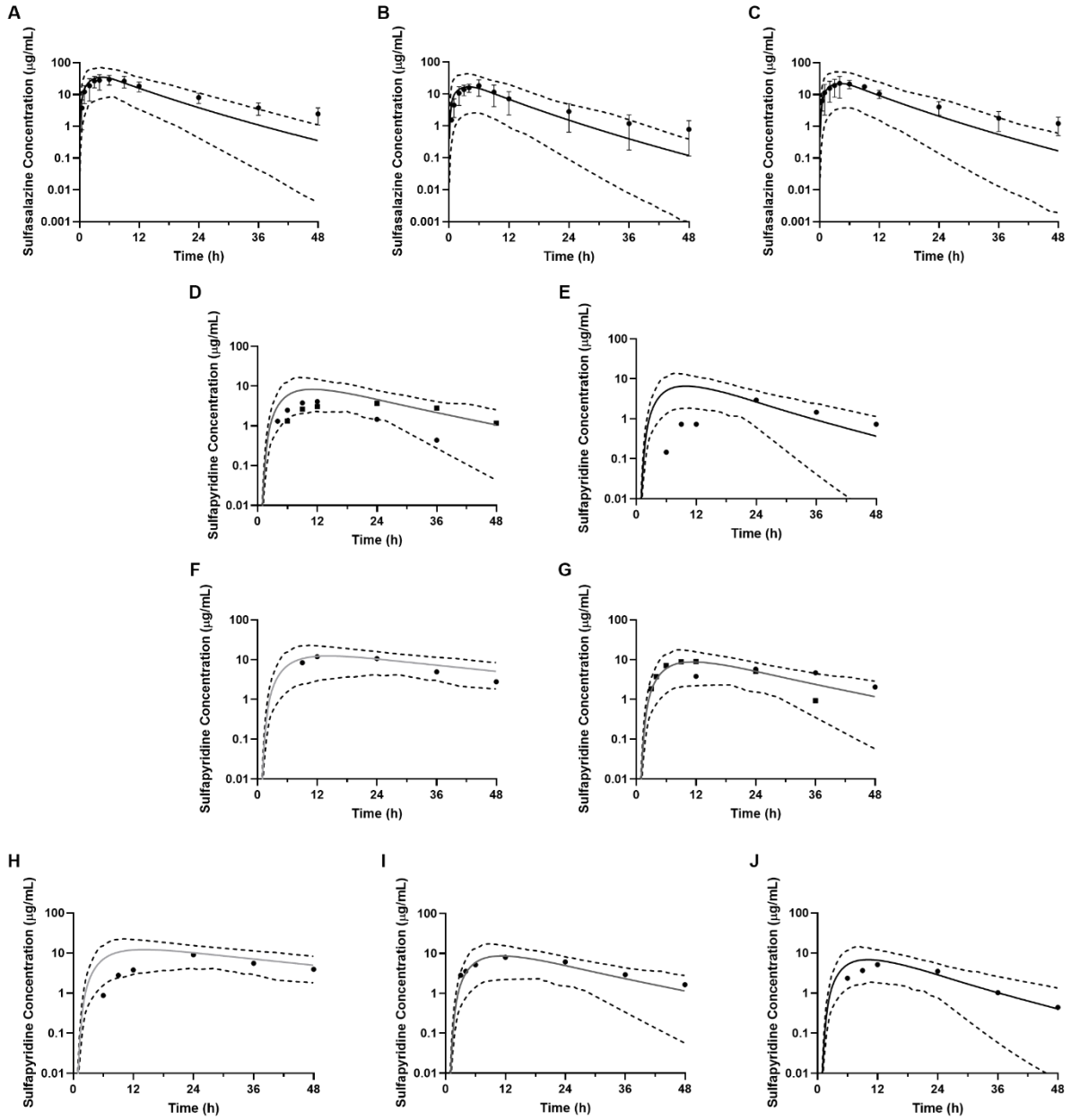


Figure S6. Observed and simulated concentration-time profiles of sulfasalazine and sulfapyridine when orally dosed with 2 g sulfasalazine plotted. Observed values are plotted as closed circles or closed squares with standard deviation where applicable. Simulated curves are represented as the solid black line for sulfasalazine. In the sulfapyridine plots, simulated mean concentrations are drawn in light gray (slow acetylators), dark gray (intermediate acetylators), and black (fast acetylators). Dashed black lines represent the simulated 5th to 95th percentile of the virtual

population. Healthy Japanese volunteers were genotyped for both BCRP and NAT2. Representative sulfasalazine profiles from the Yamasaki et al. 2008[2] study were identified as (A) AA genotype (n=3), (B) CC genotype (n=3), or (C) AC genotype (n=3) of BCRP with the PBPK simulated profiles. (D,E) Sulfapyridine PK for intermediate (D; n=2) and fast (E; n=1) metabolizers for AA genotype with simulated profiles from the respective sulfapyridine PBPK model. (F,G) Sulfapyridine profile vs. simulated PK in slow (F; n=1) and intermediate (G; n=2) acetylators in the CC genotype. (H-J) Sulfapyridine PK in slow (H; n=1), intermediate (I; n=1), and fast (J; n=1) acetylators vs. simulated PK in AC genotype. Data retrieved from Yamasaki et al. 2008[2].

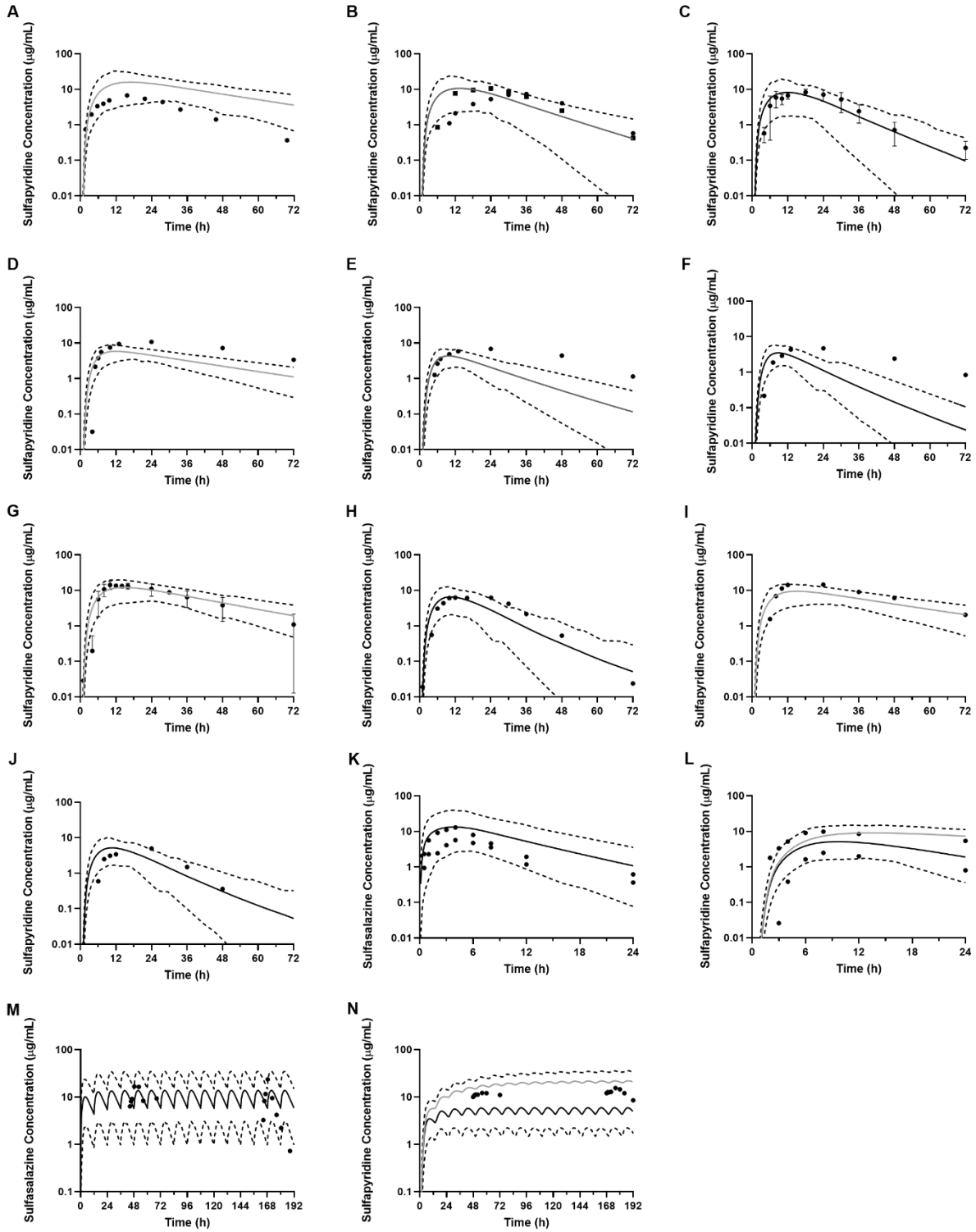


Figure S7. Observed and simulated concentration-time profiles of sulfasalazine and sulfapyridine when dosed with sulfasalazine. Observed values are plotted as closed circles or closed squares,

where studies with n=3 patients have error bars representing standard deviation. Simulated curves are represented as the solid black line for sulfasalazine. In the sulfapyridine plots, simulated mean concentrations are drawn in light gray (slow acetylators), dark gray (intermediate acetylators), and black (fast acetylators). Dashed black lines represent the simulated 5th to 95th percentile of the virtual population unless stated otherwise. (A-C) Healthy Japanese volunteers were genotyped as a (A) slow metabolizer (n=1), (B) intermediate metabolizer (n=2), and (C) fast metabolizer (n=3) were given 3 g of oral sulfasalazine (Yokogawa et al. 2001[7]). The PBPK model simulated sulfapyridine PK for each of the NAT2 genotypes. (D-F) Volunteers were genotyped for NAT2, dosed with 1 g sulfasalazine (n=6 for each acetylator type), and sulfapyridine concentration was monitored for 72 h. (Ma et al. 2009[9]) PK simulations with a virtual population matching the reported clinical study were reported for the slow, intermediate, and fast acetylators. (G-H) Observed sulfapyridine PK in slow metabolizers (n=3) and a fast metabolizer (n=1) with the corresponding PBPK simulation. (Pieniaszek et al. 1979[10]) (I-J) Bates et al. 1977[11] dosed 2 g sulfasalazine in a slow (n=1) and a fast (n=1) metabolizer plotted with the corresponding PBPK simulation. (K-L) Bugge et al. 1990[12] gave an oral dose of 1.5 g sulfasalazine to a volunteer of unknown acetylator profile. Upper dashed line in (L) is the 95th percentile of the simulated slow acetylator population and the lower dashed line shows the 5th percentile of the simulated fast acetylator population. (M-N) Sulfasalazine (n=14) and sulfapyridine (n=6) concentrations were measured on day 3 and day 8 in the Awni et al. 1995[13] study. Upper dashed line in (L) is the 95th percentile of the simulated slow acetylator population and the lower dashed line shows the 5th percentile of the simulated fast acetylator population.

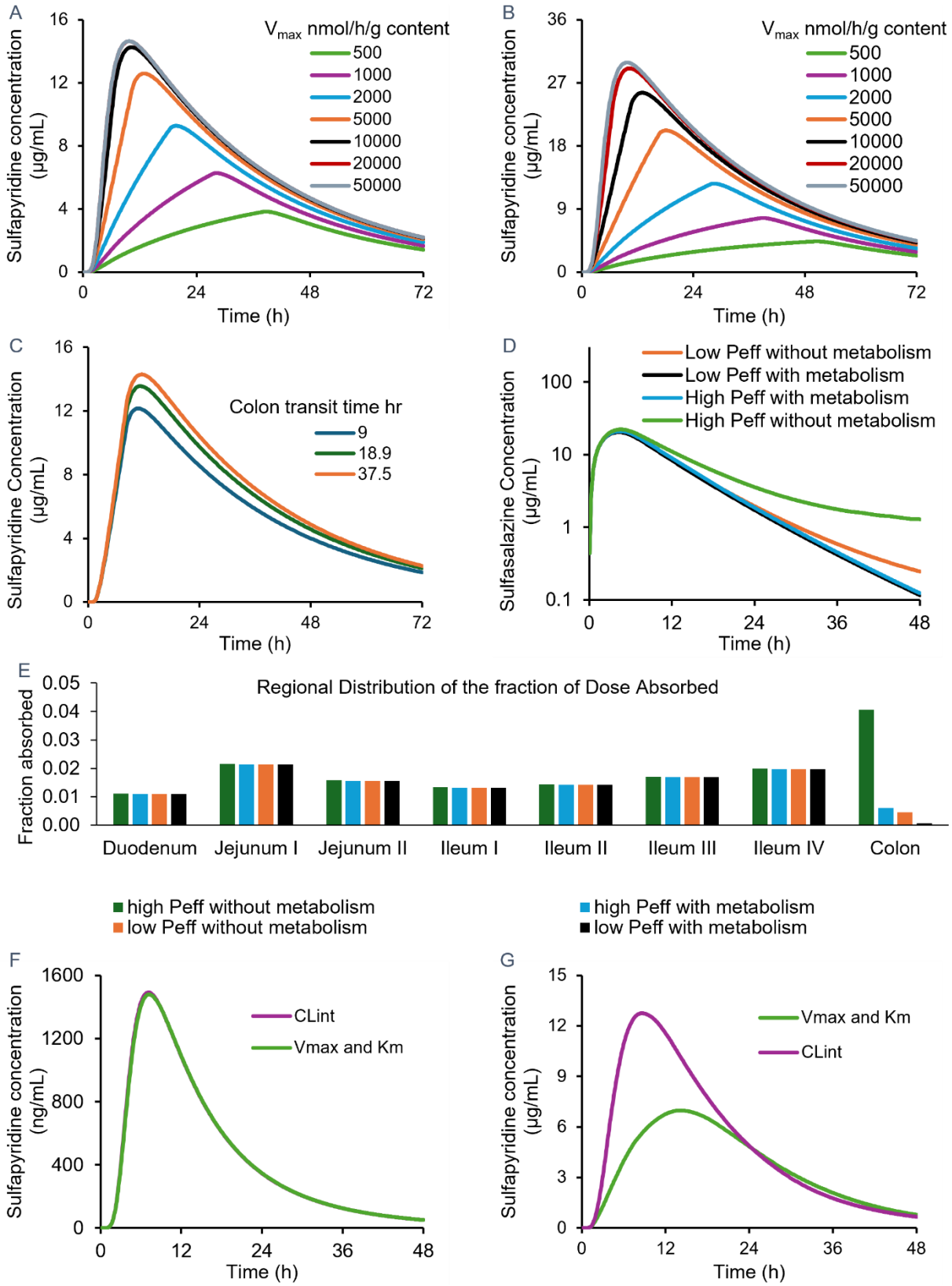


Figure S8. Additional sensitivity analysis of sulfapyridine or sulfasalazine PK after dosing

sulfasalazine to different PBPK parameters. Simulated sulfapyridine PK in population representative after dosing (A) 2 g or (B) 4 g of sulfasalazine with a range of maximum luminal metabolism rate (V_{max}). (C) Simulated sulfapyridine PK in population representative after an oral dose of 2 g sulfasalazine with various colon transit time. (D) Mean simulated sulfasalazine PK and (E) regional distribution of fraction of dose absorbed after 2 g sulfasalazine dosing with different combinations of colon effective permeability (P_{eff}) and with or without luminal drug metabolism. High P_{eff}, 0.32×10^{-4} cm/s, same as the jejunum value; low P_{eff}, 0.032×10^{-4} cm/s, a P_{eff} value 10-fold lower than that of the jejunum used in the final PBPK model to ensure negligible colon absorption; with metabolism, colon luminal metabolism enabled in simulation; without metabolism, no colon luminal metabolism in simulation.

References

1. Kusuhara H, Furuie H, Inano A, et al. Pharmacokinetic interaction study of sulphasalazine in healthy subjects and the impact of curcumin as an in vivo inhibitor of BCRP. *Br J Pharmacol*. Jul 2012;166(6):1793-803. doi:10.1111/j.1476-5381.2012.01887.x
2. Yamasaki Y, Ieiri I, Kusuhara H, et al. Pharmacogenetic characterization of sulfasalazine disposition based on NAT2 and ABCG2 (BCRP) gene polymorphisms in humans. *Clin Pharmacol Ther*. Jul 2008;84(1):95-103. doi:10.1038/sj.clpt.6100459
3. Das KM, Chowdhury JR, Zapp B, Fara JW. Small bowel absorption of sulfasalazine and its hepatic metabolism in human beings, cats, and rats. *Gastroenterology*. Aug 1979;77(2):280-4.
4. Vree TB, O'Reilly WJ, Hekster YA, Damsma JE, van der Kleijn E. Determination of the acetylator phenotype and pharmacokinetics of some sulphonamides in man. *Clin Pharmacokinet*. May-Jun 1980;5(3):274-94. doi:10.2165/00003088-198005030-00006
5. Norlander B, Gotthard R, Ström M. Pharmacokinetics of a 5-aminosalicylic acid enteric-coated tablet and suppository dosage form. *Aliment Pharmacol Ther*. Aug 1989;3(4):333-42. doi:10.1111/j.1365-2036.1989.tb00220.x
6. Yu DK, Elvin AT, Morrill B, et al. Effect of food coadministration on 5-aminosalicylic acid oral suspension bioavailability. *Clin Pharmacol Ther*. Jul 1990;48(1):26-33. doi:10.1038/clpt.1990.113
7. Yokogawa K, Nakaharu T, Ishizaki J, et al. Kinetic phenotypic diagnosis of N-acetylation polymorphism in patients based on ratio of urinary metabolites of salicylazosulfapyridine. *Int J Pharm*. Oct 23 2001;229(1-2):183-91. doi:10.1016/s0378-5173(01)00864-x
8. Schröder H, Campbell DE. Absorption, metabolism, and excretion of salicylazosulfapyridine in man. *Clin Pharmacol Ther*. Jul-Aug 1972;13(4):539-51. doi:10.1002/cpt1972134539
9. Ma JJ, Liu CG, Li JH, Cao XM, Sun SL, Yao X. Effects of NAT2 polymorphism on SASP pharmacokinetics in Chinese population. *Clin Chim Acta*. Sep 2009;407(1-2):30-5. doi:10.1016/j.cca.2009.06.025
10. Pieniaszek HJ, Jr., Resetarits DE, Wilferth WW, Blumenthal HP, Bates TR. Relative systemic availability of sulfapyridine from commercial enteric-coated and uncoated sulfasalazine tablets. *J Clin Pharmacol*. Jan 1979;19(1):39-45. doi:10.1002/j.1552-4604.1979.tb01615.x
11. Bates TR, Blumenthal HP, Pieniaszek HJ, Jr. Salivary excretion and pharmacokinetics of sulfapyridine after sulfasalazine. *Clin Pharmacol Ther*. Dec 1977;22(6):917-27. doi:10.1002/cpt1977226917
12. Buggé CJ, Gautam SR, Parke LE, Mason JT, Garcia DB. Simultaneous determination of sulfasalazine and its metabolites sulfapyridine and N-acetylsulfapyridine in human serum by ion-pair high-performance liquid chromatography using a polymer-based column. *J Pharm Sci*. Dec 1990;79(12):1095-8. doi:10.1002/jps.2600791211
13. Awni WM, Braeckman RA, Locke CS, Dubé LM, Granneman GR. The influence of multiple oral doses of zileuton on the steady-state pharmacokinetics of sulfasalazine and its metabolites, sulfapyridine and N-acetylsulfapyridine. *Clin Pharmacokinet*. 1995;29 Suppl 2:98-104. doi:10.2165/00003088-199500292-00014